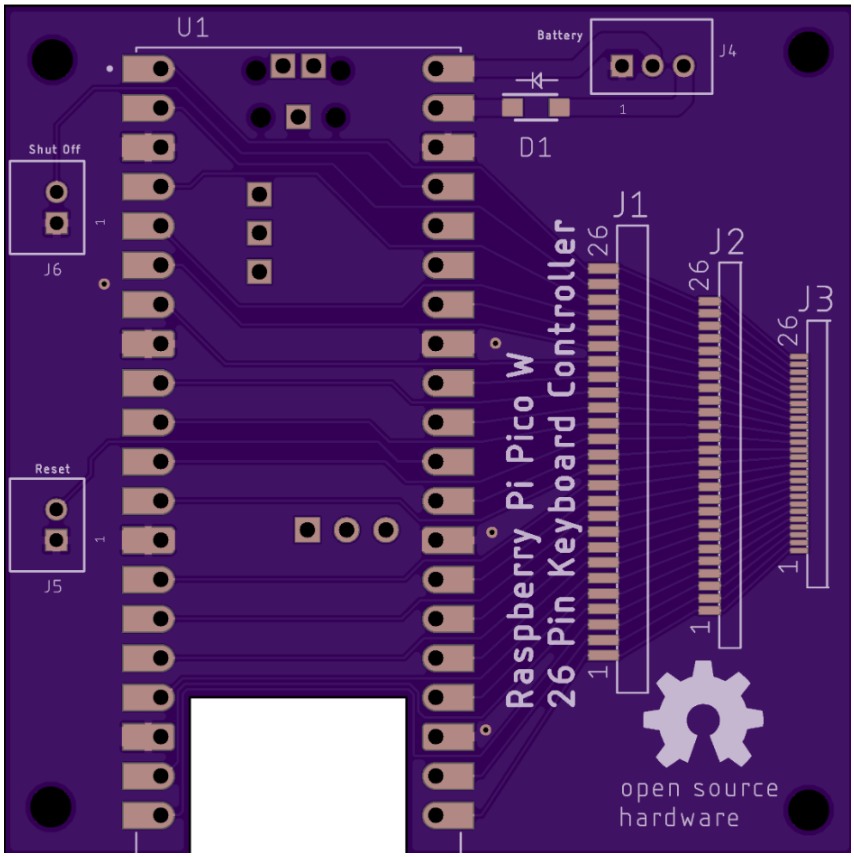
Raspberry Pi Pico W 26 Pin Keyboard Controller

This document will describe how to make a USB or BLE controller for a typical laptop keyboard with 26 or fewer FPC pins using a Raspberry Pi Pico (W) that uses the RP2040 controller IC. The USB coding can be done with Arduino C++ code, KMK, or QMK. Note that the Pi Pico 2 uses the RP2350 controller, which is not yet supported by QMK. The BLE coding is done with Arduino C++ code and will work on the Pi Pico W and the Pi Pico 2 W.

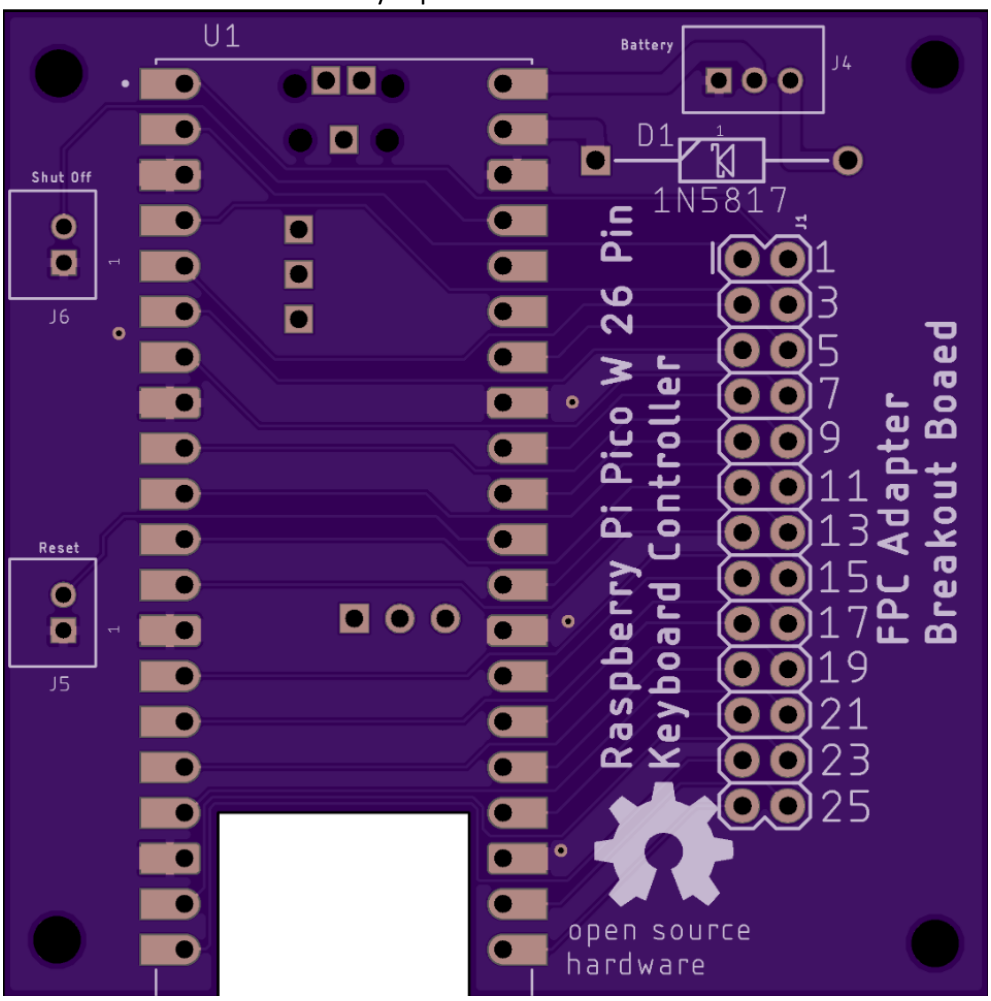
If your keyboard has more than 26 pins, you can use the RP2350 Stamp XL documented at my repo but it will only work as a KMK USB controller, not QMK. I have developed a Pi Pico (W) circuit board that controls an MCP23018 port expander for keyboards with up to 40 FPC pins. The PDF documentation for the Pico/port expander board is at my repo.

The 26 pin Pi Pico circuit board shown below has pads to solder either a 1.0mm, 0.8mm, or 0.5mm pitch FPC connector. All associated files are at my Github repository.



The cutout in the board gives better reception for the Bluetooth antenna on the Pico. The Pico can be mounted with header pins or soldered directly to the board for a lower profile. If the keyboard will only be used for USB, it will always have power. The Schottky diode and the 3 pin JST battery connector at J4 are not needed. The 2 pin JST connector at J6 is only needed if you are running under battery power and want to power down the Pico. The 2 pin JST connector at J5 is only needed if you want a reset push button switch. The "Pico_26Pin_Keyboard.brd" Eagle file and "Pico_26Pin_Keyboard.zip" Gerber file at my repo can be fabricated by OSH Park or other fab houses like JLCPCB.

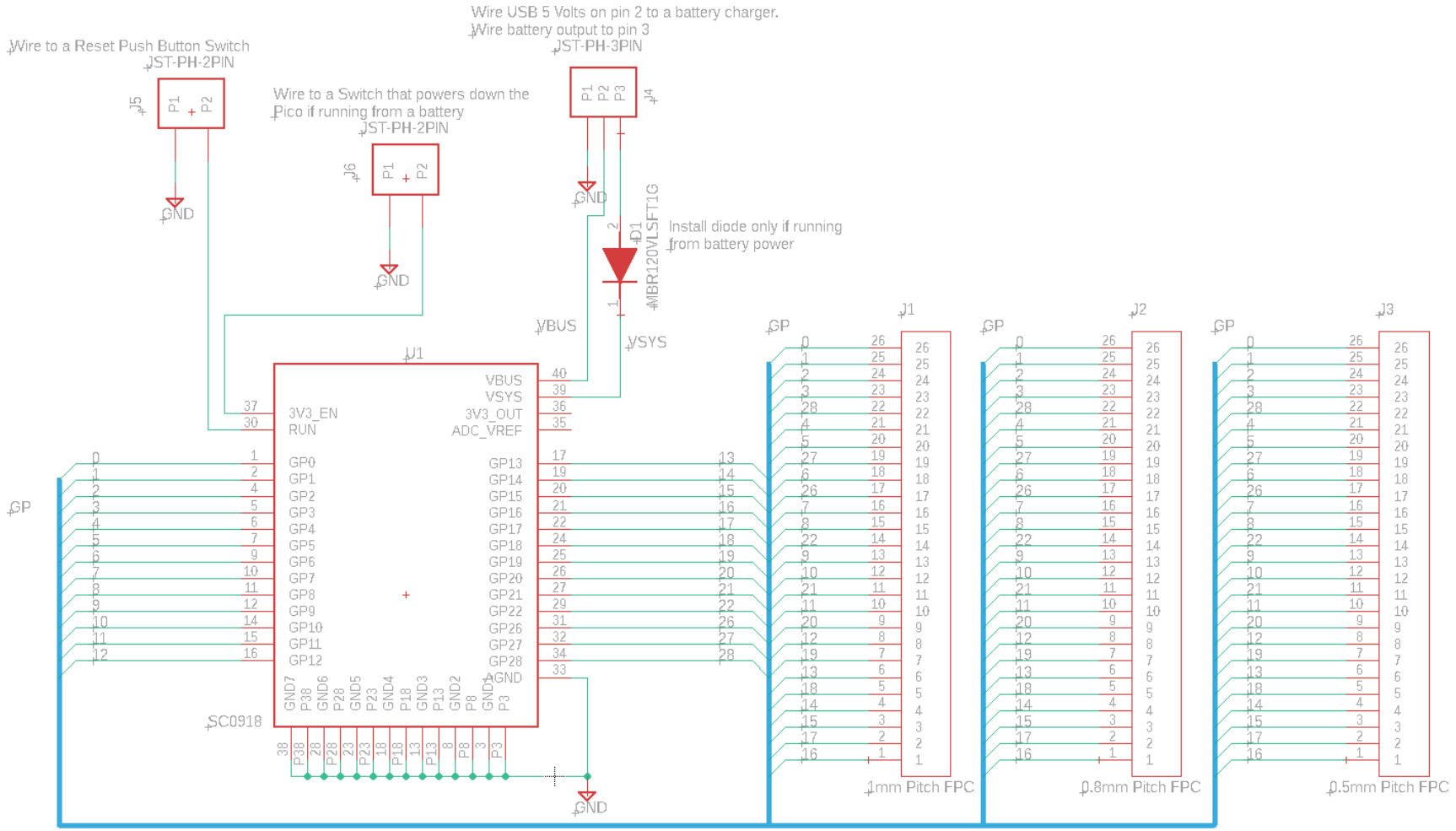
I designed a second connector board for the Pico that uses a standard FPC adapter board. Buy the adapter board without pins so you can mount it on top of my Pico board. The through hole pins on the adapter are much easier to solder compared to the fine pitch surface mount pins on the FPC connector. The traces from the Pico to the FPC connector are the same as the first board so no change is needed for the software. The "Pico_26Pin_FPC_Adapter.brd" Eagle file and "Pico_26Pin_FPC_Adapter.zip" Gerber file can be downloaded from my repo.



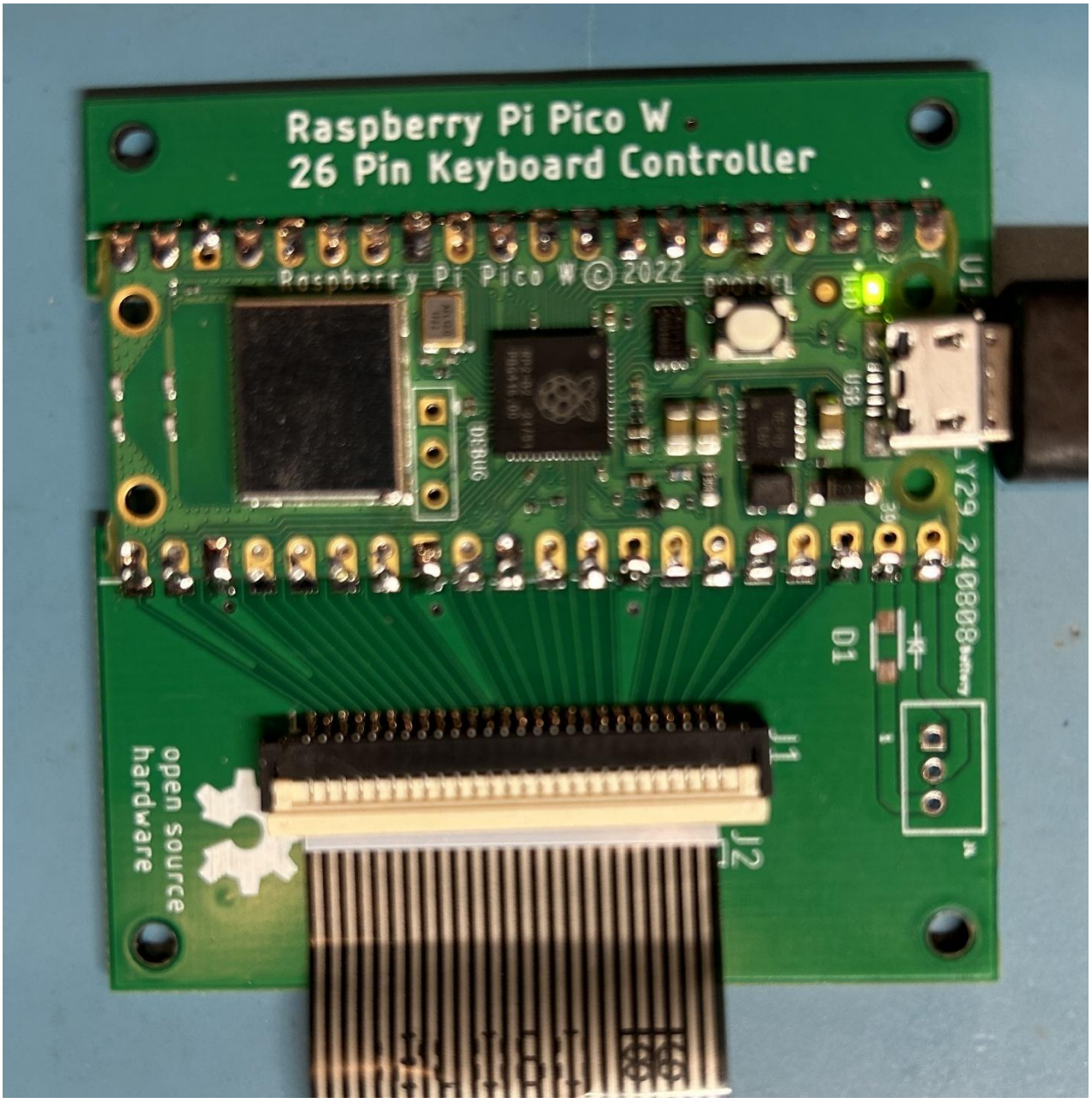
Both circuit boards measure 55mm x 55mm (2.17" x 2.17").

Bluetooth Low Energy (BLE) operation implies the controller will be running from a separate lithium battery with a charging circuit. USB 5 volts on the Pico VBUS pin is routed to pin 2 on J4 which is a 3 pin JST connector. This should be cabled (along with ground) to an off-board battery charger circuit. The nominal 3.7 volt battery output should be cabled to the JST connector J4 pin 3. This feeds a surface mount MBR120VLSFT1G or axial leaded 1N5817 Schottky diode that "or-ties" the battery voltage to the Pico's VSYS pin. There is another Schottky diode in the Pico that brings USB power to VSYS when the USB cable is attached.

The Eagle schematic "Pico_26Pin_Keyboard.sch" at my repo is shown below. The schematic for the "Pico_26Pin_FPC_Adapter" is similar (also at my repo).


Wire USB 5 Volts on pin 2 to a battery charger.
Wire battery output to pin 3
Wire to a Reset Push Button Switch
JST-PH-2PIN
JST-PH-3PIN
Wire to a Switch that powers down the Pico if running from a battery
JST-PH-2PIN
J5
J6
J4
J1
J2
J3
P1
P2
P3
GND
D1
MBR120VLSFT1G
Install diode only if running from battery power
VBUS
VSYS
U1
3V3_EN
RUN
3V3_OUT
ADC_VREF
AGND
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
GP13
GP14
GP15
GP16
GP17
GP18
GP19
GP20
GP21
GP22
GP26
GP27
GP28
GND7
P38
GND6
P28
GND5
P23
GND4
P18
GND3
P13
GND2
P8
GND1
P3
SC0918
GP
1mm Pitch FPC
0.8mm Pitch FPC
0.5mm Pitch FPC

The assembled circuit board with a Raspberry Pi Pico W and a 26 pin 1mm pitch FPC connector is shown below. I soldered this Pico directly to the connector board for a low profile but it could also be soldered with header pins. This is an early version of the board that didn't have the 2 pin JST connectors for Reset and Enable switches. The FPC cable from a Dell Inspiron 1545 keyboard is attached. The procedure to make this example keyboard operational over USB using KMK and QMK will be described first followed by the procedure to make a USB and BLE controller using Arduino C++ code.



Dell 1545 Example Keyboard Connected to the Pico



Once your circuit board has been assembled, you need to decode the internal wiring in your keyboard. This can be done with CircuitPython code or TinyUSB C++ code on the Pico. Pick the method based on which kind of keyboard controller you eventually want to make. KMK uses CircuitPython so If you want a KMK keyboard controller, you should use the CircuitPython procedure. If you want QMK, TinyUSB, or BLE, use the Arduino TinyUSB procedure. Both procedures are outlined below.

CircuitPython Keyboard Decoding Procedure:

- Connect the keyboard's FPC cable to the connector on the board with the contacts facing down.
- Python code needs an editor like Thonny which has a built-in Python interpreter that can run Python code locally on your PC, or on the Pico. Go to the bottom right corner of Thonny to select the Pico.
- To enter into boot loader mode, hold the Pico push button down while plugging in the Pico's USB cable. The new folder will show as a drive named RPI-RPI2.
- Go to the circuitpython.org/downloads page and select your Pico model. Download and copy the latest Adafruit Circuit Python uf2 over to the Pico's RPI-PPI2 folder.

- Go to circuitpython.org/libraries and download the library bundle that matches the version of Python you are using. Copy the Adafruit_HID folder into the lib folder on the Pico.
- Download the matrix_decoder_26pin.py code to your PC.
- In Thonny, select File, Open, This Computer and navigate to the code you just downloaded. This will bring it up in the editor.
- In Thonny, select the "Stop" icon. Next select File, Save As, CircuitPython Device. Overwrite the existing code.py file on the Pico with your Thonny code (renamed to code.py).
- Bring up an editor like Notepad++ on your PC and load a blank keyboard pin list with KMK key names.
- Place the PC cursor to the far right of the first key in the list (after the tabs).
- Select the "Run" icon in Thonny so the Pico will run the code.py program.
- The program will scan for any pins that are shorted together prior to any keys being pressed. The shorted pins are automatically removed from the scan list.
- Push each key on the list that exists on your keyboard. Use your PC mouse or arrow keys to skip over any keys you don't have.
- Save the completed pin list text file in your editor.

TinyUSB Keyboard Decoding Procedure:

- Connect the keyboard's FPC cable to the connector on the board with the contacts facing down.
- Load the Adafruit TinyUSB Library into the Arduino IDE.
- Set the USB stack to Adafruit TinyUSB.
- Bring up an editor like Notepad++ on your PC and load a blank keyboard pin list with QMK key names or TinyUSB key names depending on which will be your desired keyboard controller. If you eventually want to make a BLE keyboard, use the TinyUSB key names.
- Place the PC cursor to the far right of the first key in the list (after the tabs).
- Load the Pico_26Pin_Matrix_Decoder_USB.ino code into the Arduino IDE. Push and hold the button on the Pico while you plug in the USB cable. Compile and upload the code to the Pico.
- Wait for the LED on the Pico to turn on before pressing any keys. The program looks for shorted pins and removes them from the scan. You can turn on the serial monitor to see what it finds.
- Press and release each key on the list. Use your PC's arrow keys to jump over any keys on the list that your keyboard doesn't have.
- Save the completed pin list text file in your editor.

Once you have a complete connection list, there are 3 ways you can determine the row and column pins and populate the key matrix:

Method 1 - Use your favorite AI assistant (like ChatGPT, Claude, or Gemini) to analyze your filled-out keyboard pin list text file. To force the AI to accurately find the hidden matrix geometry without creating a giant, un-optimized grid, copy and paste the following KMK prompt (with noted changes for QMK or TinyUSB).

*KMK AI Prompt: "I am building a USB laptop keyboard controller using a Raspberry Pi Pico and KMK (*change to QMK or TinyUSB*). I have attached my raw keyboard GPIO pin connection list below. Because laptop matrices lack schematics and omit diodes, I do not know the exact number of rows and columns. Analyze the overlapping intersections in my pin list to isolate the optimized, condensed hardware matrix geometry. The column axis will represent a small group of shared lines (typically 8 or 9 pins), while the intersecting row axis will hold the remaining unique pins. Organize these into an efficient row_pins and col_pins configuration (do not output a giant, un-optimized square array). Finally, output a complete, KMK main.py file."*

For QMK, change the last sentence to "*Finally, output a complete, QMK keymap.c file and keyboard.json file."*

For TinyUSB, change the last sentence to "*Finally, output the row and column Pico GPIO pins and the key matrix in a two-dimensional array that can replace the base keyboard map in this Dell 1545 code."*

Here is my raw pin connection file: ***[INSERT YOUR TEXT FILE HERE or PROVIDE A LINK]****".*

Method 2 - Run my Matrix_Builder Python program in Thonny. This program runs on your PC (not on the Pico) so you'll have to change the setting in the lower right corner of Thonny. The program reads the keyboard pin connection list and outputs the row/column pins and the key matrix. This program is geared towards QMK but it will work for KMK or TinyUSB with some hand editing.

Method 3 - Use the same manual procedure described in steps 12 and 13 of my Instructable. I used this method for the Dell example keyboard to determine which GPIO's are rows and which are columns. A quick run-through using the Modifier keys; Control, Alt, Shift, GUI, and Fn is given below. Usually there will be 8 GPIO's that will be columns and programmed as Pico inputs with pull ups. The remaining GPIO's will be rows and driven low one at a time by the Pico. The remaining rows will be floated so they don't interfere. Refer to the Dell Inspiron 1545 keyboard pin connections at my repo to follow along with the methodology given below:

Control-Left and Control-Right have GPIO 8 in common which will be a row. GPIO's 2 and 27 will be columns. Shift-Left and Shift-Right have GPIO 22 in common (row) and GPIO's 2 and 5 are columns (we already knew 2 was a column). Alt-Left and Alt-Right have GPIO 9 in common (row) and GPIO's 1 and 6 are columns. So far, we know 1, 2, 5, 6, and 27 are columns. GPIO 4 on the GUI key would make sense as a column based on looking down the list. Likewise, GPIO 3 for the Fn key makes sense as a column when looking down the list. That gives 7 columns and to find the 8th, look down the list to find a key that doesn't use 1, 2, 3, 4, 5, 6, or 27. The E key shows 28 is the last column. All other GPIO's will be rows. Build a key matrix with the column GPIO's listed across the top and all the remaining GPIO's listed as rows down the side. Now place each key name at the appropriate row/column intersection as shown below.

Dell 1545 Keyboard Matrix

	GP1	GP2	GP3	GP4	GP5	GP6	GP27	GP28
GP7			PGDOWN	LGUI				PGUP
GP8		RCTRL					LCTRL	
GP9	RALT			PSCREEN		LALT		EJCT
GP10		Z	A	N1	TAB	ESC	GRAVE	Q
GP11		C	D	N3	F3	F4	F2	E
GP12	SPACE	ENTER	BSLASH	F10	BSPACE	F5	F9	
GP13		COMMA	K	N8	RBRC	F6	EQUAL	I
GP14		DOT	L	N9	F7		F8	O
GP15	LEFT			END		UP	HOME	
GP16	RIGHT			F12			INSERT	
GP17	DOWN			F11			DELETE	
GP18	SLASH		SCOLON	N0	LBRC	QUOTE	MINUS	P
GP19	N	M	J	N7	Y	H	N6	U
GP20	B	V	F	N4	T	G	N5	R
GP21		X	S	N2	CAPS		F1	W
GP22		RSHIFT			LSHIFT			
GP26			FN					

The table below is the same as above except the media keys replace the function keys when Fn is held down.

	GP1	GP2	GP3	GP4	GP5	GP6	GP27	GP28
GP7			PGDOWN	LGUI				PGUP
GP8		RCTRL					LCTRL	
GP9	RALT			PSCREEN		LALT		EJCT
GP10		Z	A	N1	TAB	ESC	GRAVE	Q
GP11		C	D	N3	F3	BRID	F2	E
GP12	SPACE	ENTER	BSLASH	MPRV	BSPACE	BRIU	VOLU	
GP13		COMMA	K	N8	RBRC	F6	EQUAL	I
GP14		DOT	L	N9	MUTE		VOLD	O
GP15	LEFT			END		UP	HOME	
GP16	RIGHT			MNXT			INSERT	
GP17	DOWN			MPLY			DELETE	
GP18	SLASH		SCOLON	N0	LBRC	QUOTE	MINUS	P
GP19	N	M	J	N7	Y	H	N6	U
GP20	B	V	F	N4	T	G	N5	R
GP21		X	S	N2	CAPS		F1	W
GP22		RSHIFT			LSHIFT			
GP26			FN					

At this point, you will create either KMK or QMK files or C++ code for a USB keyboard controller.

KMK USB Controller:

KMK is open-source firmware normally used for mechanical keyboards but it also works for laptop keyboards, as long as you know how the switches are wired (see above tables). KMK uses CircuitPython which is why I also wrote matrix decoder code in CircuitPython. All of the hard work of scanning key switches and communicating over USB is embedded in KMK. All you need to do is edit my KMK Dell example code with your key matrix information. The following assumes you already loaded the Pico with the Circuit Python library detailed in the matrix decoder section above.

Use my code_Dell1545.py as your starting point. Load it into Thonny and edit it as follows:

Change keyboard.col_pins to list the GPIO column pins left to right across the top of your matrix.

Change keyboard.row_pins to list the GPIO row pins top to bottom along the side of your matrix.

The COL2ROW setting for the keyboard.diode_orientation will force KMK to use open-drain scanning to protect diodeless matrices like those found in laptop keyboards.

Change the keyboard.keymap matrix for the base layer. Make a copy of the base layer and copy it to the Fn media layer. Add the media key name you desire at the associated function key location. Use the KMK basic key names and the media key names given at the KMK website. Each keycode name (except FN) must be preceded with KC. Put KC.NO in any cell of the matrix that has no key at that location.

After saving the code to your computer, save it to the Pico's CIRCUITPY drive with the name "code.py" (this will overwrite your previous code.py that was the matrix decoder).

In Thonny, click the "Stop" icon, then click the "Run" icon. You should now be able to bring up a blank editor like Notepad++ and type on the keyboard to test if all the keys work. If you have any typo's, they will be reported in the Thonny console window. Use the keyboardchecker.com website to verify all the function keys work.

For normal USB keyboard operation (without Thonny), you can just plug in the USB cable. This will bring up a CIRCUITPY drive folder which you can close.

QMK USB Controller: (Note QMK does not support the Raspberry Pi Pico 2 or any other RP2350 module)

QMK is widely considered the de facto open-source standard firmware for mechanical keyboards. There are lots of custom printed circuit boards for building a QMK based mechanical keyboard but very little for laptop keyboard control. The QMK C software allows up to 32 layers with advanced macros, RGB lighting, and "tap-dance" keys. I'm just a beginner when it comes to QMK but my simplified build procedure can be expanded to add many more features.

Read through the QMK tutorial and other web sites to get a feel for the build process and the command line interface (CLI). The CLI uses Python scripts so you'll need a standalone version of Python installed on your computer (not just the Python bundled inside Thonny). A good code editor like Notepad++ is also needed. Disregard talk about the browser based "QMK Configurator" as it's not useful for a laptop keyboard.

Download and run QMK-MSYS. Use its CLI to issue the command, "qmk setup". The installer will download a lot of files from GitHub including a folder full of mechanical keyboard files (that you won't use).

Use the pwd, ls, and cd Linux commands in the CLI to navigate to the following directory:

/c/Users/<user_name>

In the CLI, run the "qmk new-keyboard" command. This will create files and folders in the proper directory structure for later editing. It will ask for your new keyboard name, which can only have lower case letters, underscore, and numbers. Also, it will ask for your GitHub user name and real name, (leave these blank if you want). Select "none of the above" from the list of mechanical keyboards. Select "No" when asked if the controller is on a development board. Select the RP2040 for a Raspberry Pi Pico. Laptop keyboards typically don't have diodes but you must make the diode_direction line be COL2ROW. A new folder with your keyboard name will be created in the keyboards directory. On my Windows 11 PC, it was at:

/c/Users/<user_name>/qmk_firmware/keyboards/<keyboard_name>

Navigate to the new <keyboard_name> folder and it will have 2 files, keyboard.json and readme.md as well as one folder named keymaps. You should update the readme.md with your information for future reference. Use the Dell 1545 keyboard.json file at my repo as a guide as you edit your keyboard.json file using the data from whichever method you used to determine the row and column pins. A portion of the dell1545 keyboard.json file is shown below. The extrakey setting needs to be "true" if you are adding an Fn media key matrix. The RP2040 I/O pins are listed without quotes or GP ahead of the number. The diode_direction is defined as COL2ROW to force open-drain scanning of the rows which protects key matrices like laptops that have no diodes.

"manufacturer": "Frank Adams",

"keyboard_name": "dell1545",

"maintainer": "thedalles77",

"bootloader": "rp2040",

"diode_direction": "COL2ROW",
"features": {
"bootmagic": false, set this to true if you want to put the Pico in boot mode using the “Q” key
"extrakey": true,
"mousekey": false,
"nkro": false
},
"matrix_pins": {
"rows": [16, 17, 15, 14, 18, 13, 19, 12,
20, 11, 21, 10, 9, 22, 8, 7, 26],
"cols": [6, 27, 5, 4, 28, 3, 2, 1]
},
"processor": "RP2040",
"url": "",
"usb": {
"device_version": "1.0.0",
"pid": "0x0000",
"vid": "0xFEED"
},
"layouts": {
"LAYOUT": {
"layout": [
{"matrix": [0, 0], "x": 0, "y": 0},
{"matrix": [0, 1], "x": 1, "y": 0},
{"matrix": [0, 2], "x": 2, "y": 0},
{"matrix": [0, 3], "x": 3, "y": 0},
{"matrix": [0, 4], "x": 4, "y": 0},
{"matrix": [0, 5], "x": 5, "y": 0},
{"matrix": [0, 6], "x": 6, "y": 0},

{"matrix": [0, 7], "x": 7, "y": 0},

This pattern repeats until all keys are defined. The number of rows times the number of columns determines how many “matrix” lines are required. For the Dell 1545 example, there are 17 rows X 8 columns = 136 total. The last group of 8 pins in the keyboard.json file is given below.

{"matrix": [16, 0], "x": 0, "y": 16},

{"matrix": [16, 1], "x": 1, "y": 16},

{"matrix": [16, 2], "x": 2, "y": 16},

{"matrix": [16, 3], "x": 3, "y": 16},

{"matrix": [16, 4], "x": 4, "y": 16},

{"matrix": [16, 5], "x": 5, "y": 16},

{"matrix": [16, 6], "x": 6, "y": 16},

{"matrix": [16, 7], "x": 7, "y": 16}

Note; there is no comma after the last matrix line.

The next step is to edit the keymap.c file found two levels down at:

/c/Users/<your_user_name>/qmk_firmware/keyboards/<keyboard_name>/keymaps/default

Use the matrix table from whichever method you used above. Populate the first array with the base layer key names. The Dell 1545 keymap.c file shown below is at my repo.

```
#include QMK_KEYBOARD_H
const uint16_t PROGMEM keymaps[][MATRIX_ROWS][MATRIX_COLS] = {
    [0] = LAYOUT(
        KC_NO,   KC_INS,  KC_NO,   KC_F12,  KC_NO,   KC_NO,   KC_NO,   KC_RGHT,
        KC_NO,   KC_DEL,  KC_NO,   KC_F11,  KC_NO,   KC_NO,   KC_NO,   KC_DOWN,
        KC_UP,   KC_HOME, KC_APP,  KC_END,  KC_NO,   KC_NO,   KC_NO,   KC_LEFT,
        KC_NO,   KC_F8,   KC_F7,   KC_9,    KC_O,    KC_L,    KC_DOT,  KC_NO,
        KC_QUOT, KC_MINS, KC_LBRC, KC_0,    KC_P,    KC_SCLN, KC_NO,   KC_SLSH,
        KC_F6,   KC_EQL,  KC_RBRC, KC_8,    KC_I,    KC_K,    KC_COMM, KC_NO,
        KC_H,    KC_6,    KC_Y,    KC_7,    KC_U,    KC_J,    KC_M,    KC_N,
        KC_F5,   KC_F9,   KC_BSPC, KC_F10,  KC_NO,   KC_BSLS, KC_ENT,  KC_SPC,
        KC_G,    KC_5,    KC_T,    KC_4,    KC_R,    KC_F,    KC_V,    KC_B,
        KC_F4,   KC_F2,   KC_F3,   KC_3,    KC_E,    KC_D,    KC_C,    KC_NO,
        KC_NO,   KC_F1,   KC_CAPS, KC_2,    KC_W,    KC_S,    KC_X,    KC_NO,
        KC_ESC,  KC_GRV,  KC_TAB,  KC_1,    KC_Q,    KC_A,    KC_Z,    KC_NO,
        KC_LALT, KC_NO,   KC_NO,   KC_PSCR, KC_NO,   KC_NO,   KC_NO,   KC_RALT,
        KC_NO,   KC_NO,   KC_LSFT, KC_NO,   KC_NO,   KC_NO,   KC_RSFT, KC_NO,
        KC_NO,   KC_LCTL, KC_NO,   KC_NO,   KC_NO,   KC_NO,   KC_RCTL, KC_NO,
        KC_NO,   KC_NO,   KC_NO,   KC_LGUI, KC_PGUP, KC_PGDN, KC_NO,   KC_NO,
        KC_NO,   KC_NO,   KC_NO,   KC_NO,   KC_NO,   MO(1),   KC_NO,   KC_NO
    ),
```

The second array holds the media keys that are activated when the Fn key is held down.

```
[1] = LAYOUT(
    KC_NO,   KC_INS,  KC_NO,   KC_MNXT, KC_NO,   KC_NO,   KC_NO,   KC_RGHT,
    KC_NO,   KC_DEL,  KC_NO,   KC_F11,  KC_NO,   KC_NO,   KC_NO,   KC_DOWN,
    KC_UP,   KC_HOME, KC_APP,  KC_END,  KC_NO,   KC_NO,   KC_NO,   KC_LEFT,
    KC_NO,   KC_VOLD, KC_MUTE, KC_9,    KC_O,    KC_L,    KC_DOT,  KC_NO,
    KC_QUOT, KC_MINS, KC_LBRC, KC_0,    KC_P,    KC_SCLN, KC_NO,   KC_SLSH,
    KC_F6,   KC_EQL,  KC_RBRC, KC_8,    KC_I,    KC_K,    KC_COMM, KC_NO,
    KC_H,    KC_6,    KC_Y,    KC_7,    KC_U,    KC_J,    KC_M,    KC_N,
    KC_BRIU, KC_VOLU, KC_BSPC, KC_MPRV, KC_NO,   KC_BSLS, KC_ENT,  KC_SPC,
    KC_G,    KC_5,    KC_T,    KC_4,    KC_R,    KC_F,    KC_V,    KC_B,
    KC_BRID, KC_F2,   KC_F3,   KC_3,    KC_E,    KC_D,    KC_C,    KC_NO,
    KC_NO,   KC_F1,   KC_CAPS, KC_2,    KC_W,    KC_S,    KC_X,    KC_NO,
    KC_ESC,  KC_GRV,  KC_TAB,  KC_1,    KC_Q,    KC_A,    KC_Z,    KC_NO,
    KC_LALT, KC_NO,   KC_NO,   KC_PSCR, KC_NO,   KC_NO,   KC_NO,   KC_RALT,
    KC_NO,   KC_NO,   KC_LSFT, KC_NO,   KC_NO,   KC_NO,   KC_RSFT, KC_NO,
    KC_NO,   KC_LCTL, KC_NO,   KC_NO,   KC_NO,   KC_NO,   KC_RCTL, KC_NO,
    KC_NO,   KC_NO,   KC_NO,   KC_LGUI, KC_PGUP, KC_PGDN, KC_NO,   KC_NO,
    KC_NO,   KC_NO,   KC_NO,   KC_NO,   KC_NO,   KC_TRNS, KC_NO,   KC_NO
)
```

QMK basic keycodes start with KC_ followed by the key name. Use KC_NO at any array location that has no key. Put MO(1) at the Fn position in the base layer and KC_TRNS at the Fn position in the second layer. If you want additional layers, there are lots of tutorials that give examples.

After editing keyboard.json and keymap.c, use the CLI to navigate to /c/Users/<user_name>/qmk_firmware/keyboards/<keyboard_name>

Compile the code in the CLI with the command:

qmk compile -kb <your_keyboard_name> -km default

The compile takes 30 minutes on my slow computer (once I clear the typos and syntax errors). The QMK source code is updated every 3 months so depending on what version you downloaded from their repo, you may need to change the syntax of the commands shown here.

Flashing your controller:

To load your new QMK code into a Raspberry Pi Pico, start with the USB disconnected, push and hold the button on the Pico while plugging in the USB cable (or the “Q” key if you enabled bootmagic). Release the button and a folder will show up as RPI-RPI2 with a drive letter. Drag the <keyboard_name>_default.uf2 file that was just compiled into the Pico folder. The .uf2 file is located at C:Users/<user_name>/qmk_firmware

The RPI-RPI2 folder will blink out and the keyboard should be functional. Unlike KMK, there is no folder that opens whenever you plug in the USB cable.

Arduino C++ USB Code

You should have already loaded the Adafruit TinyUSB Library into the Arduino IDE from the earlier step of decoding the keyboard. Use the information you gathered from building your key matrix to modify the example USB code for a Dell 1545 laptop keyboard. The sections to modify are given below.

const int NUM_ROWS = 17; // number of row signals

const int NUM_COLS = 8; // number of column signals

// column and row GPIO numbers attached to the column and row pins of the keyboard

const int colPins[NUM_COLS] = {1, 2, 3, 4, 5, 6, 27, 28};

const int rowPins[NUM_ROWS] = {7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 26};

```
const uint8_t keyMap[NUM_ROWS][NUM_COLS] = {
  { 0,                  0,                    HID_KEY_PAGE_DOWN,  HID_KEY_GUI_LEFT,     0,                    0,                  0,                    HID_KEY_PAGE_UP },
  { 0,                  HID_KEY_CONTROL_RIGHT,0,                  0,                    0,                    0,                  HID_KEY_CONTROL_LEFT, 0 },
  { HID_KEY_ALT_RIGHT,  0,                    0,                  HID_KEY_PRINT_SCREEN, 0,                    HID_KEY_ALT_LEFT,   0,                    0 },
  { 0,                  HID_KEY_Z,            HID_KEY_A,          HID_KEY_1,            HID_KEY_TAB,          HID_KEY_ESCAPE,     HID_KEY_GRAVE,        HID_KEY_Q },
  { 0,                  HID_KEY_C,            HID_KEY_D,          HID_KEY_3,            HID_KEY_F3,           HID_KEY_F4,         HID_KEY_F2,           HID_KEY_E },
  { HID_KEY_SPACE,      HID_KEY_ENTER,        HID_KEY_BACKSLASH,  HID_KEY_F10,          HID_KEY_BACKSPACE,    HID_KEY_F5,         HID_KEY_F9,           0 },
  { 0,                  HID_KEY_COMMA,        HID_KEY_K,          HID_KEY_8,            HID_KEY_BRACKET_RIGHT,HID_KEY_F6,         HID_KEY_EQUAL,        HID_KEY_I },
  { 0,                  HID_KEY_PERIOD,       HID_KEY_L,          HID_KEY_9,            HID_KEY_F7,           0,                  HID_KEY_F8,           HID_KEY_O },
  { HID_KEY_ARROW_LEFT, 0,                    0,                  HID_KEY_END,          0,                    HID_KEY_ARROW_UP,   HID_KEY_HOME,         0 },
  { HID_KEY_ARROW_RIGHT,0,                    0,                  HID_KEY_F12,          0,                    0,                  HID_KEY_INSERT,       0 },
  { HID_KEY_ARROW_DOWN, 0,                    0,                  HID_KEY_F11,          0,                    0,                  HID_KEY_DELETE,       0 },
  { HID_KEY_SLASH,      0,                    HID_KEY_SEMICOLON,  HID_KEY_0,            HID_KEY_BRACKET_LEFT, HID_KEY_APOSTROPHE, HID_KEY_MINUS,        HID_KEY_P },
  { HID_KEY_N,          HID_KEY_M,            HID_KEY_J,          HID_KEY_7,            HID_KEY_Y,            HID_KEY_H,          HID_KEY_6,            HID_KEY_U },
  { HID_KEY_B,          HID_KEY_V,            HID_KEY_F,          HID_KEY_4,            HID_KEY_T,            HID_KEY_G,          HID_KEY_5,            HID_KEY_R },
  { 0,                  HID_KEY_X,            HID_KEY_S,          HID_KEY_2,            HID_KEY_CAPS_LOCK,    0,                  HID_KEY_F1,           HID_KEY_W },
  { 0,                  HID_KEY_SHIFT_RIGHT,  0,                  0,                    HID_KEY_SHIFT_LEFT,   0,                  0,                    0 },
  { 0,                  0,                    0,                  0,                    0,                    0,                  0,                    0 }
};
```

The above key matrix uses the key names required by the Adafruit TinyUSB library. The Fn - media keys were not coded.

Bluetooth Low Energy (BLE)

KMK and QMK do not support BLE on the Pi Pico W but it is supported by the Earle Philhower Arduino C++ libraries. Debugging Bluetooth can be frustrating so I recommend you get your new keyboard working first over USB using Adafruit TinyUSB (see previous section). Then you can modify the pins and key matrix in my BLE Arduino code for the Dell 1545 keyboard per the following steps:

You will need to load the Earle Philhower C++ BLE libraries into the Arduino IDE. Under Tools, USB Stack, select Pico SDK. Also under tools, IP/Bluetooth Stack, select IPv4 + Bluetooth.

Each time you re-flash the Pico, you need to “Remove Device" in your computer or phone's Bluetooth settings and then pair it again.

The sections you will need to change are shown below.

```
const int NUM_ROWS = 17; // number of row signals
const int NUM_COLS = 8; // number of column signals

// column and row GPIO numbers attached to the column and row pins of the keyboard
const int colPins[NUM_COLS] = {1, 2, 3, 4, 5, 6, 27, 28};
const int rowPins[NUM_ROWS] = {7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 26};

// Base Keyboard Map. A zero indicates no key at that position.
const uint8_t keyMap[NUM_ROWS][NUM_COLS] = {
  { 0,              0,              KEY_PAGE_DOWN,  KEY_LEFT_GUI,     0,              0,              0,              KEY_PAGE_UP },
  { 0,              KEY_RIGHT_CTRL, 0,              0,                0,              0,              KEY_LEFT_CTRL,  0 },
  { KEY_RIGHT_ALT,  0,              0,              KEY_PRINT_SCREEN, 0,              KEY_LEFT_ALT,   0,              0 },
  { 0,              'z',            'a',            '1',              KEY_TAB,        KEY_ESC,        '`',            'q' },
  { 0,              'c',            'd',            '3',              KEY_F3,         KEY_F4,         KEY_F2,         'e' },
  { ' ',            KEY_RETURN,     '\\',           KEY_F10,          KEY_BACKSPACE,  KEY_F5,         KEY_F9,         0 },
  { 0,              ',',            'k',            '8',              ']',            KEY_F6,         '=',            'i' },
  { 0,              '.',            'l',            '9',              KEY_F7,         0,              KEY_F8,         'o' },
  { KEY_LEFT_ARROW, 0,              0,              KEY_END,          0,              KEY_UP_ARROW,   KEY_HOME,       0 },
  { KEY_RIGHT_ARROW,0,              0,              KEY_F12,          0,              0,              KEY_INSERT,     0 },
  { KEY_DOWN_ARROW, 0,              0,              KEY_F11,          0,              0,              KEY_DELETE,     0 },
  { '/',            0,              ';',            '0',              '[',            '\'',           '-',            'p' },
  { 'n',            'm',            'j',            '7',              'y',            'h',            '6',            'u' },
  { 'b',            'v',            'f',            '4',              't',            'g',            '5',            'r' },
  { 0,              'x',            's',            '2',              KEY_CAPS_LOCK,  0,              KEY_F1,         'w' },
  { 0,              KEY_RIGHT_SHIFT,0,              0,                KEY_LEFT_SHIFT, 0,              0,              0 },
  { 0,              0,              0,              0,                0,              0,              0,              0 }
};
```

Put a 0 in any location that has no key. Use the names above as guides for what you need to put in your new matrix. Note the letters, numbers, and punctuation are inside single quotes but other keys use KEY_xxx. Backspace and quote are special:

‘\\’ this is the backspace key

‘\’’ this is the single quote key

Change the Bluetooth name associated with your keyboard at line 57:

KeyboardBLE.begin("Dell1545 Keyboard", "Dell Mod");

The Fn - media keys were not coded.